

Core Java Interview Questions and Answers (50 Questions)

Topics Covered:

- Collections
- Lambda Expressions
- Streams API
- String
- Exception Handling
- Optional
- Records
- Multithreading (Basic)
- JDBC (Basic)

Suitable for 0–2 years experience.

String Questions

1. Why String is immutable in Java?

Answer

String objects are immutable to improve security, thread safety, and performance. Once created, their value cannot be changed. Immutability also helps Java maintain the String Pool efficiently.

2. Difference between String, StringBuilder, and StringBuffer?

Answer

String is immutable, while StringBuilder and StringBuffer are mutable. StringBuilder is faster but not thread-safe, whereas StringBuffer is synchronized and thread-safe.

3. What is String Pool?

Answer

String Pool is a special memory area in the heap where Java stores string literals. It avoids duplicate object creation and improves memory efficiency.

4. Difference between == and equals() in String?

Answer

== compares object references, while equals() compares actual content. For comparing string values, equals() should be used.

5. Why String class is final?

Answer

String is final to prevent inheritance and maintain immutability. This improves security, reliability, and consistent behavior.

6. What is substring() method?

Answer

substring() is used to extract a portion of a string based on index positions. It returns a new string without modifying the original string.

7. What is the difference between charAt() and substring()?

Answer

charAt() returns a single character from a string, while substring() returns a portion of the string. Both are commonly used for string manipulation.

Collections Questions

8. What is Collection Framework?

Answer

The Collection Framework is a set of classes and interfaces used to store and manipulate groups of objects. It provides reusable data structures and algorithms.

9. Difference between List, Set, and Map?

Answer

List stores ordered duplicate elements, Set stores unique elements, and Map stores key-value pairs. Each collection type is used for different scenarios.

10. Difference between ArrayList and LinkedList?

Answer

ArrayList uses a dynamic array and provides fast random access. LinkedList uses nodes and performs better for insertions and deletions.

11. Difference between HashSet and TreeSet?

Answer

HashSet stores unordered unique elements using hashing. TreeSet stores sorted unique elements using a tree structure.

12. Difference between HashMap and Hashtable?

Answer

HashMap is not synchronized and allows null values, while Hashtable is synchronized and does not allow null keys or values.

13. Difference between HashMap and ConcurrentHashMap?

Answer

HashMap is not thread-safe, while ConcurrentHashMap supports concurrent access with better performance in multithreaded applications.

14. How HashMap works internally?

Answer

HashMap uses hashing to store key-value pairs in buckets. It uses hashCode() and equals() methods to locate keys efficiently.

15. What is load factor in HashMap?

Answer

Load factor determines when HashMap should resize its internal bucket array. The default load factor is 0.75.

16. What is fail-fast iterator?

Answer

Fail-fast iterators throw ConcurrentModificationException if the collection is modified while iterating. They immediately stop iteration to prevent inconsistent behavior. Iterators from ArrayList and HashMap are common examples of fail-fast iterators.

17. Difference between Comparable and Comparator?

Answer

Comparable is used for natural sorting and requires implementing `compareTo()` inside the class itself. Comparator is used for custom sorting logic and is implemented separately using `compare()`. Comparator is useful when multiple sorting strategies are needed.

Answer

Comparable provides natural sorting using `compareTo()`, while Comparator provides custom sorting using `compare()`.

18. What is Queue?

Answer

Queue is a collection interface that follows FIFO (First In First Out) order. Elements are inserted at the rear and removed from the front. Queues are commonly used in scheduling systems, task processing, and messaging systems.

19. What is Stack?

Answer

Stack is a collection that follows LIFO (Last In First Out) order. The last inserted element is removed first. Stack is commonly used in recursion, undo operations, and expression evaluation.

Answer

Stack follows LIFO (Last In First Out) order. It is commonly used in recursion and undo operations.

Exception Handling Questions

20. What is exception handling?

Answer

Exception handling is a mechanism used to handle runtime errors gracefully without terminating the program abruptly. Java uses try, catch, finally, throw, and throws keywords for exception handling. Proper exception handling improves application reliability.

21. Difference between checked and unchecked exceptions?

Answer

Checked exceptions are verified at compile time and must be handled using try-catch or throws. Unchecked exceptions occur at runtime and are subclasses of RuntimeException. Examples include IOException for checked and NullPointerException for unchecked exceptions.

Answer

Checked exceptions are checked at compile time and must be handled explicitly. Unchecked exceptions occur at runtime.

22. Difference between throw and throws?

Answer

throw is used to explicitly throw an exception object, while throws is used in method declarations to indicate possible exceptions.

23. What is finally block?

Answer

finally block always executes whether an exception occurs or not. It is mainly used for cleanup activities such as closing database connections, files, or streams. finally improves resource management and application stability.

24. What is custom exception?

Answer

Custom exceptions are user-defined exceptions created to represent business-specific errors. They are usually created by extending Exception or RuntimeException classes. Custom exceptions improve readability and better error handling.

Answer

Custom exceptions are user-defined exceptions created for business-specific error handling.

25. What is try-with-resources?

Answer

try-with-resources automatically closes resources such as streams and database connections after execution.

Lambda Expression Questions

26. What is lambda expression?

Answer

Lambda expression provides a concise way to represent anonymous functions in Java. It reduces boilerplate code and supports functional programming.

Example:

`(a, b) -> a + b`

27. What is functional interface?

Answer

A functional interface contains exactly one abstract method. It is mainly used with lambda expressions.

28. What are common functional interfaces?

Answer

Common functional interfaces include Predicate, Function, Consumer, and Supplier. They are available in `java.util.function` package.

29. Difference between anonymous class and lambda expression?

Answer

Lambda expressions provide shorter and cleaner syntax compared to anonymous classes. Lambdas are mainly used for functional interfaces.

30. What is Predicate interface?

Answer

Predicate is a functional interface that accepts one input and returns a boolean result. It is mainly used for conditional checks and filtering operations in streams. Predicate contains the `test()` method.

31. What is Function interface?

Answer

Function is a functional interface that accepts one input and returns a transformed output. It is commonly used in `map()` operations within streams. Function contains the `apply()` method.

Answer

Function interface accepts one input and returns one output after transformation.

32. What is Consumer interface?

Answer

Consumer is a functional interface that accepts input and performs operations without returning any result. It is commonly used for printing, logging, and updating data. Consumer contains the `accept()` method.

33. What is Supplier interface?

Answer

Supplier is a functional interface that provides data without taking any input parameters. It is mainly used for lazy object creation and generating values dynamically. Supplier contains the `get()` method.

Answer

Supplier interface provides data without taking any input.

Streams API Questions

34. What is Stream API?

Answer

Stream API is used for processing collections declaratively. It supports filtering, mapping, sorting, and aggregation operations.

35. Difference between Collection and Stream?

Answer

Collections store data in memory, while Streams process data dynamically. Streams support functional-style operations.

36. What is filter() in streams?

Answer

filter() is an intermediate stream operation used to select elements based on conditions. It returns a new stream containing only matching elements. filter() is commonly used for search and conditional data processing.

37. What is map() in streams?

Answer

map() is an intermediate stream operation used to transform elements into another form. It applies a function to every element and returns a new stream. map() is commonly used for converting objects or extracting fields.

Answer

map() transforms each element into another form and returns a new stream.

38. Difference between map() and flatMap()?

Answer

map() transforms elements one-to-one, while flatMap() flattens nested structures into a single stream.

39. What is collect() in streams?

Answer

collect() is a terminal stream operation used to gather processed stream results into collections or other data structures. It is commonly used with Collectors.toList(), toSet(), and groupingBy(). collect() is one of the most frequently used stream operations.

40. What is forEach() in streams?

Answer

forEach() is a terminal operation used to iterate through stream elements one by one. It is commonly used for printing or performing actions on elements. forEach() does not return a new stream.

Answer

forEach() is a terminal operation used to iterate through stream elements.

41. Difference between intermediate and terminal operations?

Answer

Intermediate operations return another stream and are lazy, while terminal operations produce final results.

42. What is parallel stream?

Answer

Parallel streams process stream elements concurrently using multiple threads from the ForkJoinPool framework. They help improve performance for large datasets and CPU-intensive operations. However, parallel streams should be used carefully because thread management overhead may affect small datasets.

43. What is Optional in Java?

Answer

Optional is a container object introduced in Java 8 to avoid NullPointerException. It represents values that may or may not be present and provides methods like `isPresent()`, `orElse()`, and `ifPresent()`. Optional improves null handling and code readability.

Answer

Optional is a container object introduced in Java 8 to avoid NullPointerException. It represents values that may or may not be present.

44. What is Optional.ofNullable()?

Answer

`Optional.ofNullable()` creates an Optional that can hold either null or non-null values safely.

45. What is record in Java?

Answer

Record is a special Java class introduced to reduce boilerplate code for immutable data objects. It automatically generates constructors, getters, `equals()`, `hashCode()`, and `toString()` methods. Records are commonly used for DTOs and simple data carriers.

46. What is multithreading?

Answer

Multithreading allows multiple threads to execute concurrently within a single application. It improves application performance, responsiveness, and resource utilization. Java supports multithreading using Thread class and Executor framework.

Answer

Multithreading allows multiple threads to execute concurrently within a program. It improves performance and responsiveness.

47. What is synchronization?

Answer

Synchronization is used to control access to shared resources in multithreaded applications. It prevents race conditions and ensures data consistency when multiple threads access the same data. Synchronization can be applied at method or block level.

48. What is JDBC?

Answer

JDBC stands for Java Database Connectivity and is an API used to connect Java applications with relational databases. It allows executing SQL queries, retrieving results, and performing database operations. JDBC is widely used in enterprise applications.

Answer

JDBC stands for Java Database Connectivity. It is used to connect Java applications with databases and execute SQL queries.

49. Difference between Statement and PreparedStatement?

Answer

Statement is used for executing static SQL queries, while PreparedStatement supports parameterized queries. PreparedStatement improves performance by precompiling queries and also prevents SQL injection attacks. It is preferred in real-world applications.

50. What is garbage collection?

Answer

Garbage Collection is the JVM process that automatically removes unused objects from memory. It helps manage memory efficiently and reduces the risk of memory leaks. Java provides automatic memory management using different garbage collection algorithms.

Answer

Garbage Collection automatically removes unused objects from memory. It helps manage memory efficiently and prevents memory leaks.